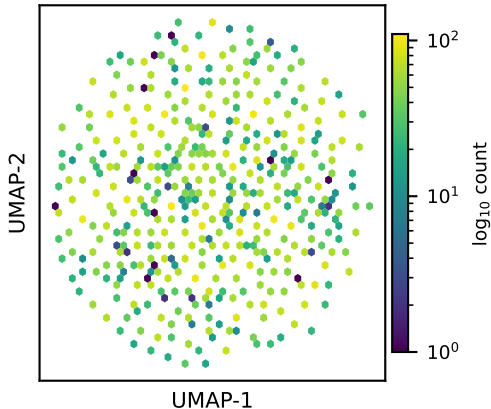
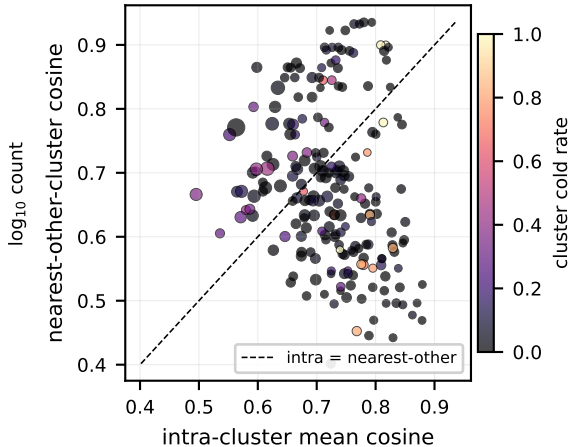


(A) ECAPA UMAP manifold  
(19,101 chunks, log density)



(B) Per-cluster tightness vs  
separation (k=210, marker  $\propto$  size)



(C) raw-L2 192-D validation

**Validated: SHIPPED k=210 labels**  
(production cluster.py space;  
NOT the optimistic UMAP-32)

**kNN label cohesion (k=10)**  
0.694 vs chance 0.0074  
= 94x (load-bearing)

**HDBSCAN (independent)**  
406 clusters, noise 3.8%  
non-degenerate; NMI 0.77  
(ARI 0.31: 406>210 granularity)

Agglomerative@210 NMI 0.76  
Silhouette 0.073 (relative only)